Isometric Standard Project Verification Report

For

Charm Industrial

Client	Isometric
Company / GHG Removal Supplier	Charm Industrial
Facility or Project Name	Charm Kansas Bio-Oil Geologic Storage
Facility Address	**Bio-oil Production:** **AE Cote-Nord Canada Bioenergy Inc (AECN)** Port Cartier, Quebec, Canada **Charm Miniforge** Fort Lupton, CO, USA **Bio-Oil Pre-Processing:** **Charm Industrial** El Dorado, KS, USA **Bio-oil Injection:** **Vaulted Deep** Hutchinson, KS, USA
Greenhouse Gas Statement ID	Isometric ggs_1JXX6BQZ31S0XMSE
Net Greenhouse Gas Removals	1014.57 tonnes
Removal Method	Bio-Oil Geological Storage
Removal Period	Apr 1, 2025 – May 31, 2025
Verification Opinion	Fully meets criteria
Verifier/Validation and Verification Body	Tim Hansen 350Solutions, Inc.

ISSUED: JULY4, 2025

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ISOMETRIC STANDARD VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: *Max Lavine	Validation and Verification Team: *Tim Hansen, PE Guy Ingram Hardwick
Removal Method: Bio-Oil Geological Storage		
Report Date: July 4, 2025		
Document No: 350VR-Charm-ISO2504.04		
Rev: v1.3		

** primary contact/lead author*

1. KEY PROJECT DATA

TABLE 1. CHARM TECHNOLOGY VERIFICATION SUMMARY

Title	Charm Kansas Bio-Oil Geologic Storage
Location	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Charm Industrial Miniforge 2360 S Rollie Ave Fort Lupton, CO, USA 80621 40.076570, -104.804680 Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 37.813906, -96.914982 Bio-oil Injection: Vaulted Deep 7513 KS-14 Hutchinson, KS 67501 37.966035, -97.947845
Reporting Period	April 1, 2025 – May 31, 2025
Greenhouse Gas Statement ID	Isometric ggs_1JXX6BQZ31S0XMSE (Statement ID) Isometric gss_1JY4KBETX1S07ZVE (Version ID)
Project Proponents	CDR Supplier: Charm Industrial Feedstock Supplier: Produits Forestiers Arbec S.E.N.C (Quebec mill waste supplier); Altitude Forestry (CO forestry fire prevention waste supplier) Geologic Storage Management: Vaulted Deep Regulatory Agency (Injections / Storage): Kansas Department of Health and Environment (KDHE)
Legal Ownership of the Credits	Charm Industrial
Removals in Reporting Period	1014.57 tonnes
Removal Method	Bio-Oil Geological Storage in Salt Caverns

Verification Report Date	July 4, 2025
Verification Type & Criteria	Isometric Standard Verification for ongoing deliveries of carbon removal credits by Charm Industrial in accordance with the documents used during project validation: • Isometric Standard v1.7 • Bio-Oil Geological Storage Protocol v1.1 • Biomass Feedstock Accounting Module v1.2 • Biomass or Bio-oil Storage in Salt Caverns v1.1 • Embodied Emissions Accounting v 1.0 • Energy Use Accounting v 1.2 • Transportation Emissions Accounting v1.1

1.1. VERIFICATION SUMMARY

350Solutions was contracted to perform a project verification of the greenhouse gas (GHG) statement of Charm Industrial's bio-oil geological storage process for reporting period including April 1, 2025 – May 31, 2025. The validation report for the Charm Kansas Bio-oil Geologic Storage project was completed on June 25, 2024, by Futurepast, which focused on the use of third-party produced bio-oils and was issued by Isometric for the Crediting Period March 1, 2024 – February 28, 2029. An additional validation of the Charm Kansas Bio-oil Geologic Storage project for use of bio-oil produced by Charm was completed on October 18, 2024, for the same crediting period. These validations remain valid for this verification reporting period. Further details and findings can be found in the validation report (https://registry.isometric.com/project/prj_1HZSSWBQM1S08H83#project-documentation).

350Solutions affirms that Charm has the appropriate equipment, procedures, and protocols in place and documented to quantify GHG removal through the injection of bio-oil into salt caverns via permitted Class V wells in accordance with the requirements of the applicable Isometric standards, protocols and modules.

The carbon removals for the reporting period are provided in Table 2. These values have been verified by 350Solutions.

TABLE 2. VERIFIED TOTAL REMOVALS BY CHARM INDUSTRIAL FOR REPORTING PERIOD ENDING MAY 31, 2025

Removal ID	Supplier reference ID	Date removal started	Date removal completed	CO_2e net removed (kg)
rmv_1JXNT8KSP1S01NC2	844	2025-05-13	2025-05-13	39320
rmv_1JXNSPNNF1S0Y9S9	840	2025-05-02	2025-05-02	49480
rmv_1JXNSG51Q1S0RN94	838	2025-04-29	2025-04-29	44020
rmv_1JXNS70WH1S0FMK3	836	2025-04-25	2025-04-25	40940
rmv_1JXNTTHAZ1S0T0X0	824	2025-05-20	2025-05-20	49410
rmv_1JXNTCVH71S0JZYY	845	2025-05-15	2025-05-15	40450
rmv_1JXNT3KEK1S0A8TF	843	2025-05-12	2025-05-12	41950
rmv_1JXNV4HM61S06K1A	827	2025-05-29	2025-05-29	41930
rmv_1JXNRQKG01S0K0ZX	833	2025-04-18	2025-04-18	48980

rmv_1JXNRKVS91S0XQG6	832	2025-04-17	2025-04-17	41800
rmv_1JXNSDAES1S07AP8	837	2025-04-28	2025-04-28	44900
rmv_1JXNT0H3R1S065MF	842	2025-05-09	2025-05-09	32220
rmv_1JXNTPSF61S0GKQN	847	2025-05-19	2025-05-19	39100
rmv_1JXNS2NEZ1S09Z3P	835	2025-04-24	2025-04-24	47480
rmv_1JXNV1B741S061WP	826	2025-05-27	2025-05-27	43080
rmv_1JXNSM21J1S0TPCG	839	2025-05-01	2025-05-01	51850
rmv_1JXNRZBYA1S0RMGT	834	2025-04-21	2025-04-21	45120
rmv_1JXNR0WD41S05RTQ	829	2025-04-03	2025-04-03	40410
rmv_1JXNYHY6F1S08W94	828	2025-04-01	2025-04-01	20790
rmv_1JXNSVE1Z1S00NDF	841	2025-05-05	2025-05-05	40850
rmv_1JXNRF9F41S0WSW8	831	2025-04-15	2025-04-15	52840
rmv_1JXNTKRXY1S0N12A	846	2025-05-16	2025-05-16	47190
rmv_1JXNTXPQQ1S0W78S	825	2025-05-22	2025-05-22	48190
rmv_1JXNRBDRB1S0JDGM	830	2025-04-14	2025-04-14	22270
TOTAL				1014570

The quantity of 1014.57 tonnes of net carbon removals has been verified with a reasonable level of assurance and are eligible for crediting under the referenced Isometric Standard and Protocol.

350Solutions Inc. declares that it is an impartial verification body, approved by Isometric for verifications in the required technical area, free from any conflicts of interest, capable, and qualified to complete this verification for the current reporting period according to the *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2.1*, *Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*.

Verifier Information		
Verification Body	**Verifier**	**Verification ID No.**
350Solutions, Inc.	Tim Hansen	VR-Charm-ISO2504.04

Timothy A. Hansen Digitally signed by Timothy A. Hansen Date: 2025.07.04 07:37:22 -04'00'

Signed: Tim Hansen (Verifier)

Guy Hardwick Digitally signed by Guy Hardwick Date: 2025.07.04 13:19:52 +01'00'

Guy Ingram-Hardwick (QA)

2. VERIFICATION DETAILS

2.1. VERIFICATION OBJECTIVES

A series of verification activities was conducted by 350Solutions to independently verify the claimed removals of the GHG statement submitted by Charm Industrial for operations conducted at the Kansas Bio-oil Geologic Storage facility for the Reporting Period April1, 2025 – May 31, 2025. The verification was conducted following the specifications of *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2*, *Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*. The verification is based on the provision and fair presentation of a complete and accurate GHG Statement by Charm Industrial and Isometric. This is confirmed and verified by 350Solutions based on supporting evidence in accordance with the Isometric criteria (Table 1, Section 3.2).

2.2. LEVEL OF ASSURANCE & MATERIALITY

The verification was conducted with a reasonable level of assurance. Any discrepancies between the claimed removals of the project and the removals estimated by the verification process for the reporting period are less than the materiality threshold of 5%.

2.3. VERIFICATION TEAM DETAILS

350Solutions is an accredited inspection & verification body by ANAB under ISO 17020:2012 for completion of ISO 14034:2016 Technology Verifications. 350Solutions is headquartered in Raleigh, North Carolina, USA. 350Solutions Technical Lead for the Charm verification is Tim Hansen. Mr. Hansen is a registered Professional Engineer with over 20 years of experience in the evaluation and verification of novel cleantech and carbon technologies. Mr. Hansen has led verifications of carbon removal technologies for multiple registries, including Isometric, and has experience in the development and testing of biomass conversion technologies, including gasification and pyrolysis. The verification report was provided QA by Guy Ingram-Hardwick, a Carbon Removal Verification Engineer, with experience developing MRV requirements for and verifying carbon removal technologies.

2.4. AUDIT PROCESS FOLLOWED

The specific verification activities conducted are summarized in Table 3. The list of all files that were reviewed is provided in Appendix 1. The list of findings from the verification can be found in Appendix 2. Audit results are documented using 350's Compliance Checklist, which includes all requirements from the Isometric standard, Bio-oil Geologic storage protocol, and referenced modules.

TABLE 3. VERIFICATION ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
June 16 - June 30, 2025	Review of reporting period modifications and changes	- Review of GHG Statement and report and guidance from Charm	V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx Apri_May 2025 Removals Isometric GHG Statement Report v1.0.docx (1) Verifier Guidance -- April_May 2025 ggs_1JXX6BQZ31S0XMSE

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
June 16 – June 30, 2025	Review of current Validation and GHG Statement Applicability	- Review of Validation Report - Review of prior Verification Report - Identification of major and minor process steps, measurements, and calculations and changes in RP - Review of Standards/Protocols/Modules required for Verification - Generation of a checklist of verification issues and findings and opportunities for improvement	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1 V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx Apri_May 2025 Removals Isometric GHG Statement Report v1.0.docx (1) Verifier Guidance -- April_May 2025 ggs_1JXX6BQZ31S0XMSE Isometric Standard V1.7.2, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2, Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1, and Energy Use Accounting v1.2
June 16 – June 30, 2025	Verification of Removals	- Verify changes to the system - Review of measurements and calculations - Review of durability/permit requirement monitoring data - Review of GHG statement - Identification of initial findings and follow-up request for information - Write verification report	File List provided in Appendix 1. Initial Findings in Appendix 2.

3. TECHNICAL ASPECTS

3.1. PROJECT DETAILS

Charm obtained bio-oil from two sources during this reporting period. Charm obtains bio-oil from a producer in Quebec, Canada (AECN) who uses an Ensyn/ Honeywell UOP pyrolysis system to generate bio-oil from waste wood produced at a neighboring mill. In addition, Charm produces bio-oil at its pyrolysis facility in Fort Lupton, CO. Production of bio-oil for sequestration and carbon removal credit generation began in August of 2024, with the initial installation of 4 pyrolyzers, with plans for up to 18 pyrolyzers total to be installed. Charm's pyrolyzers utilize wood chips, placed into a high temperature pressurized reactor in a low oxygen environment to reduce the wood chips into a combination of products, including syngas and light hydrocarbon gases, pyrolysis oils, biochar, and wood vinegar. All hot pyrolysis gases from the reactor are quenched with water to produce the oil and wood vinegar products. Excess gases are routed to a combustion device for emissions control, with the heat produced from the combustion of excess gas utilized for drying the biomass from 40%-50% moisture content to 10% moisture content before introduction into the pyrolyzer.

Bio-oil from both facilities is shipped by tanker truck or rail car to a Charm facility in El Dorado, Kansas. At this facility, Charm pretreats the bio-oil through the addition and sparging of waste brine (salt) and liquid caustic soda (LCS) to adjust the pH of the bio-oil and the salt content or conductivity to meet the requirements of the injection facility permit for storage in salt caverns as specified by the Kansas Department of Health and Environment (KDHE) and injection partner Vaulted Deep's Class V injection permit. Following treatment, samples are taken for CHN analysis (carbon content), total acid number (TAN), and density testing and the oil is transported to the Vaulted Deep injection facility.

Once successfully pre-treated to ensure EPA Underground Injection Control (UIC)/KDHE compliance, including measurement of pH and conductivity both within allowable permitted ranges, the oil is trucked to Vaulted Deep's Great Plains Facility in Hutchinson, KS for injection into an underground salt cavern. At that point, the oil is offloaded and Vaulted takes custody of the product and responsibility for emplacing it into the storage cavern. The waste is not stored at the Vaulted site, instead it is processed and injected on the same day as it arrives on site. Therefore, emissions associated with waste storage at the facility are avoided. Vaulted is also contracted to monitor the emplacement cavern and provide Charm with MRV data from their monitoring equipment. The details of Vaulted's injection infrastructure and monitoring equipment are provided in their own PDD, which has been previously validated [1] and remains valid for the reporting period.

3.2. PROTOCOLS AND MODULES APPLIED

The verification was conducted following the specifications of the following standards, protocols, and modules:

- Isometric Standard v1.7 [2]
- Bio-Oil Geological Storage Protocol v1.1 [3]
- Biomass Feedstock Accounting Module v1.2 [4]
- Biomass or Bio-oil Storage in Salt Caverns v1.1 [5]
- Embodied Emissions Accounting v 1.0 [6]
- Energy Use Accounting v 1.2 [7]
- Transportation Emissions Accounting v 1.1 [8]

Charm utilizes Book and Claim Units (BCUs) to substitute for some, or all, or project diesel fuel usage and purchases Renewable Energy Certificates (RECs) for some or all of its electricity consumption.

3.3. APPLICABILITY OF APPLIED PROTOCOL AND MODULES

The Charm project takes mixed woody waste or forestry thinnings from wildfire prevention activities, converts it to bio-oil, which they pretreat and inject into salt caverns. The biomass feedstocks were assessed previously under the eligibility criteria of the Isometric Biomass Feedstock Module and were found to be eligible. As the biomass feedstocks have been deemed eligible biomass feedstocks, the storage of biomass conversion products (bio-oil) in underground salt caverns makes both the Bio-oil Geological Storage Protocol and Biomass/Bio-oil Storage in Salt Caverns Module applicable to this project. All other modules for energy, transportation, and embodied emissions accounting are generally applicable to this project.

3.4. PROJECT BOUNDARY

The boundaries of the project include the supply of wood waste to the biomass processing sites (AECN and Charm Fort Lupton), the conversion of the biomass to bio-oil via AECN's pyrolysis process and Charm's pyrolysis process, transport of the bio-oil to Charm for preprocessing at its El Dorado, Kansas facility, completion of pretreatment, shipment to the Vaulted Deep Great Plains facility, and injection of the bio-oil in the salt caverns. Monitoring and required analysis of the bio-oil and the storage formation are also included in the project boundary.

3.5. CHANGES FROM PREVIOUS VERIFICATIONS

Compared to prior verifications, Charm has made modifications to the process, data collection, or calculation procedures which have been identified in the GHG statement and supporting documentation. The following modifications in project activities were made, in addition to updates to delivery data and monitoring data for the reporting period:

- Charm is now shipping both the bio-oil fraction of its liquid products as well as the Quench Oil / Wood Vinegar fraction to its processing facility for future injections once approved by KDHE. This fraction is not yet being injected, however.

3.6. PROCESS & DATA IRREGULARITIES DURING REPORTING PERIOD

Several anomalies and irregularities occurred during the reporting period related to data collection and operations. All have been reviewed and the approach Charm has taken for each accepted and determined to be compliant with protocols. The following issues were identified by Charm and reviewed by 350Solutions:

- Electricity usage during one injection batch was recorded incorrectly. Additional evidence was provided (photos of meter readings) and the usage updated to match meter readings.
- For one delivery, the truck did not arrive on time for injection, so had to be rerouted and return the next day. Charm accounted for all associated transportation emissions appropriately.
- For multiple salt deliveries to the El Dorado facility, delivery weigh scale tickets or bills of lading showing delivered salt weights were not provided by drivers. Charm used conservative approaches to account for delivery weights and the associated transportation emissions. Actual salt usage is documented clearly in operating logs.

4. CREDITING DETAILS

4.1. PROJECT REPORTING PERIOD

This verification considers injections of bio-oil by Charm during the period April 1, 2025 to May 31, 2025. The reporting period is based on the actual injection dates, with bio-oil production, transportation, monitoring, and related activities potentially occurring outside of this reporting period, but properly accounted for and allocated to the bio-oil injected within the reporting period.

4.2. CREDITING PERIOD

The 5-year crediting period for the Charm Kansas Bio-oil Geologic Sequestration Project began March 1, 2024, and continues for five years, ending February 28, 2028. All claimed removals in this reporting period fall within the current crediting period.

5. GHG STATEMENT VERIFICATION

5.1. ACCURACY OF THE STATEMENT

The values represented in the GHG statement have been evaluated, cross-referenced with the presented evidence, and confirmed. Confirmation of the GHG statement has been based on the following efforts:

- Review and confirmation of calculations of total bio-oil and CO_2 equivalent injected and all process, operational, and embodied emissions, as well as all supporting evidence for the following removals

rmv_1JXNYHY6F1S08W94	828	2025-04-01
rmv_1JXNTXPQQ1S0W78S	825	2025-05-22
rmv_1JXNRBDRB1S0JDGM	830	2025-04-14

- This includes for these removals:
    - Calculation of mass of bio-oil delivered and injected including tracing to raw data (weigh scale tickets, carbon content, addition of consumables (salt and LCS))
    - Review and calculation of average carbon content from raw data (lab reports)
    - Confirmation of transportation distances for tanker and rail car deliveries from AECN to El Dorado, KS
    - Confirmation of weigh scale raw data for deliveries of bio-oil and injectate
    - Re-calculation of transportation associated emissions for such transport
    - Confirmation of raw data for fuel usage at Charm pre-treatment facility
    - Confirmation of electricity usage at Charm pretreatment facility and Vaulted Deep injection facility from raw data
    - Confirmation of proper use and accounting of BCUs and RECs to offset diesel and electricity usage emissions, respectively
    - Review of Charm pyrolysis process emissions, supporting evidence, and calculations
- Review of monitoring data as required for salt cavern injection / emplacement and post injection monitoring to ensure ongoing compliance with Class V permit.

As with the previous verification, some minor discrepancies were noted due to rounding errors both internally at Charm and in communication between Charm's and Isometric's data platforms. The estimated impact of the total discrepancies is not a material issue. No other impacts on the claimed net removals value were identified.

5.2. VERIFICATION OPINION

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the greenhouse gas removal statement indicates that the claims fall within the 5% materiality threshold for greenhouse gas emissions and removals as required by the Isometric Standard:

☒ ***Meets the requirements of the Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Meets the requirements with <u>minor revisions</u> of Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Does Not Meet the requirements of the Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

5.3. VERIFICATION RESULTS SUMMARY

A summary of verification results and compliance with relevant requirements of the *Isometric Standard, Biomass Geological Storage Protocol, Biomass Feedstock Accounting Module*, and *Biomass or Bio-oil Storage in Salt Caverns Module* are summarized below.

TABLE 4. VERIFICATION RESULTS

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1.	*Validated against approved Protocols*	N	Acceptable – No change since validation
2.5.1	*Boundaries*	N	Acceptable – No change since validation
2.5.2	*Baseline*	N	Acceptable – No change since validation
2.5.3	*Additionality - Financial*	N	Acceptable – No change since validation
2.5.3	*Additionality - Environmental*	N	Acceptable – No change since validation
2.5.3	*Additionality - Regulatory*	N	Acceptable – No change since validation
2.5.4	*Leakage*	N	Acceptable – No change since validation
2.5.5	*Default Emission Factors, Proxies, and Models*	N	Acceptable – No additional significant changes were made since prior verifications. See Section 3.6 for description of irregularities, noting that all were properly and conservatively addressed in LCA net removal calculations.
2.5.11	*Life Cycle Assessment (LCA) Policies*	N	
2.5.6	*Common Calculation Factors*	N	
2.5.7	*Uncertainty*	N	
2.5.8	*Durability and Monitoring*	N	Acceptable – No change since validation
2.5.9	*Risk of Reversal*	N	Acceptable – No change since validation
2.5.10 ***3.7***	*Environmental and Socioeconomic Impacts*	N	Acceptable – No change since validation.
3.1	*Ownership*	N	Acceptable – No change since validation
3.2	*Documentation*	Y	Acceptable – No changes since prior verification. All documentation properly updated for reporting period
3.3	*Eligibility*	Y (GHG Statement Updated)	Acceptable – Project registered with Isometric and reporting period GHG Statement properly submitted to Isometric dMRV Certify platform.
3.4	*Project Crediting*	Y (GHG Statement Updated)	Acceptable – New reporting period GHG Statement is complete and all removals have been verified. Reporting period is within current validation crediting period.
3.5	*Stakeholder Input Process*	N	Acceptable – No change since validation
3.6	*Regulatory Compliance*	Y (Required Monitoring Updated)	Acceptable – Maintenance of well operation permits meets requirements and ongoing permit reporting requirements completed. Note that a displaced brine TOC sample was not collected in May due to construction at the site and inability to access brine sampling locations. Isometric has allowed an exception for this reporting period.

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
Biomass Feedstock Accounting Module			
2.1	*Eligibility Criteria for Market Leakage*	N	Acceptable – No change since validation
2.2	*Counterfactual Storage Eligibility*	N	Acceptable – No change since validation
2.3	*Dedicated Energy Feedstock Eligibility*	N	Acceptable – No change since validation
3.0	*Counterfactual Definition*	N	Acceptable – No change since validation
3.1	*Counterfactual Energy Definition*	N	Acceptable – No change since validation
3.2	*Counterfactual Replacement Definition*	N	Acceptable – No change since validation
Biomass or Bio-oil Storage in Salt Caverns Module			
1.0, 2.2	*Salt Cavern Characterization – Cavern Dimensions*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Integrity*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Porosity & permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Subsurface Features*	N	Acceptable – No change since validation
1.0	*Salt Cavern Characterization – Temperature, pH, conductivity/Cl- conc.*	Y (Required Monitoring Updated)	Acceptable - All required parameters measured and are within the required limits.
1.0	*Salt Cavern Characterization – Brine TOC*	Y (Required Monitoring Updated)	Acceptable – Brine TOC measured prior to emplacement, as required by Class V permit. Confirmed via updated TOC data for April. Note that a displaced brine TOC sample was not collected in May due to construction at the site and inability to access brine sampling locations. Isometric has allowed an exception for this reporting period.
1.0	*Salt Cavern Characterization – $d^{13}C$ analysis (where applicable)*	NA	Acceptable – No change since validation
1.0, 3.1.3	*Salt Cavern Characterization – Surface Subsistence*	NA	Acceptable - No change since validation
1.0	*Salt Cavern Characterization – Displaced Brine Injection location*	NA	Acceptable - No change since validation

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
1.0, 2.3	*Emplacement Infrastructure*	N	Acceptable – No change since validation
1.0	*Emplacement Monitoring*	Y (Required Monitoring Updated)	Acceptable – monitoring completed and updated as required by Class V well permit (April and May monthly reports and Quarterly groundwater monitoring report to KDHE provided)
1.0, 3.2	*Well Closure and Post Closure Requirements*	N	Acceptable – No change since validation
2.1	*Permit for Operation*	N	Acceptable – No change since validation
3.1.1	*Injection Pressure*	Y (Required Monitoring Updated)	Acceptable – Pressure is continuously monitored and reported to KDHE monthly. Updates provided as required.
3.1.1	*Volume monitoring*	Y (Required Monitoring Updated)	Acceptable – Monitoring of injected material measured daily and reported monthly, confirmed by sonar quarterly. Updates provided as required.
3.1.1	*Injectate monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required. Note that a displaced brine TOC sample was not collected in May due to construction at the site and inability to access brine sampling locations. Isometric has allowed an exception for this reporting period.
3.1.1	*Sampling Frequency*	N	Acceptable – No change since validation. All sampling data provided and updated as required
3.1.1	*Safety Monitoring*	N	Acceptable – No change since validation
3.1.2	*Mechanical Integrity*	N	Acceptable – No change since validation
3.1.3	*Well Pressure Monitoring*	Y (Required Monitoring Updated)	Acceptable – Well pressure measured continuously with safety interlock. Daily pressure reported to KDHE monthly. Updated monitoring data provided as required.
3.1.3	*Sonar Survey*	N	Acceptable – No change since validation
3.1.3	*Fill Volume*	Y (Required Monitoring Updated)	Acceptable – Cavern volume tracked based upon volume of emplaced material. Estimate volume reported monthly, verified by sonar quarterly. Updated monitoring data provided as required.
3.1.3	*Return Brine Monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required. Note that a displaced brine TOC sample was not collected in May due to construction at the site and inability to access brine sampling locations. Isometric has allowed an exception for this reporting period.
3.1.3	*Volatile Gas Monitoring*	N	Acceptable – No change since validation
3.3	*Reversal Risk*	N	Acceptable – No change since validation
3.3.1	*Attribution of Reversals*	N	Acceptable – No change since validation

5.4. VERIFICATION FINDINGS

5.4.1. CRITICAL FINDINGS AND EXCEPTIONS

Assessment of the greenhouse gas removal statement identified several initial findings (Appendix 2). All findings, primarily associated with missing supporting evidence, have been addressed and closed. Findings of note include the following:

Type	Finding / Issue	Required Action	Supplier Response	Resolution
Non-Compliance (Omission)	The Monthly Vaulted KOLAR report to KDHE for May 2025 was not provided	Please provide a copy of the report	A copy of the report has been received from Vaulted and submitted	Complete – May report provided
Non-Compliance (Omission)	The May TOC sample results for displaced brine were not provided	Charm has indicated that the sample results are taken at end of month and may not be available until after the GHG Statement is submitted and will be submitted in next RP. 350 to discuss with Isometric. Or, Charm can provide results if available at this time	Vaulted has communicated that they were unable to perform scheduled testing due to site construction leaving wells inaccessible.	Complete – Isometric exception provided for this reporting period
Omission	As stated in Charm's Verifier Guidance document, Batch 2-86 electricity usage at Vaulted was noted as incorrect in Vaulted records and was corrected based on photo evidence provided to Charm by Vaulted	Please provide the photo evidence for the Vaulted meter	Photos have been provided in the "Verification Feedback" folder in the shared Drive	Complete – evidence provided
Omission / Forward Action Request	Lack of data obtained for salt deliveries as noted in Charms Verifier Guidance document provides less certainty in associated emissions and is an indicator of potential data tracking and management issues. It is noted that the approach taken to fill data gaps appears conservative and overall does not materially effect the calculations of net removals	Current approach for data gap filling is acceptable. Please work with suppliers to ensure proper weigh tickets / BOL are provided for each delivery to ensure accuracy. Please consider improvements to data collection.	We will continue working with the team on site to better manage delivery drivers and improve scaling consistency	Complete for current verification - conservative approach utilized. Review at future verifications

An exception to protocol requirements was confirmed by Isometric regarding the missing May, 2025 displaced brine TOC sample results on June 30th 2025 (Isometric email June 30, 2025 and Slack Message on July 2, 2025):

Isometric has provided the project an allowance to proceed with their verification in this case however, going forward:

- *The displaced brine TOC data must be submitted immediately upon availability;*
- *The registry reserves the right to pause verification proceedings for missing emplaced and displaced brine TOC data in future;*
- *The absence of data will be clearly documented as a deviation in the Verification Report;*
- *Any necessary discounts due to reversal risk may be made retroactively based on receipt of results;*
- *A clear plan is submitted to prevent recurrence of this issue in future periods. The plan should outline:*
    - *Early sampling in the event of planned site closure.*
    - *Collection of a statistically representative number of replacement samples in the event of damaged/ missing/ failed samples for analysis.*

5.4.2. FORWARD ACTION REQUESTS & RECOMMENDATIONS

Forward Action requests (FAR) and recommendations are also provided in Appendix 2 for consideration for improvement of future operations and GHG Removal Credit calculations and to ensure tracking of potential issues to be monitored by verifiers in future verifications. Open Forward Action Requests and opportunities for improvement are summarized below for future verification reference:

TABLE 5. FORWARD ACTION REQUESTS AND RECOMMENDATIONS

Type	Finding / Issue	Required Action	Supplier Response	
Forward Action Request	350 is aware that storage partner Vaulted is completing some equipment and facility upgrades. If applicable to Charm injection processes, the embodied emissions should be included in Charm LCA.	Please verify if Vaulted has implemented any equipment upgrades and additional equipment related to Charm's injections and ensure the Vaulted embodied emissions factor is updated if so.	Understood, will confirm with Vaulted prior to the next reporting	Review at future verifications
Forward Action Request	Non-pyrolysis electricity consumption (G&A, Eng) is based on records from May 2024. Please evaluate	Please update values in next Reporting Period or provide evidence that this is a	Absolutely, we will revisit this during the next Reporting Period	Review at future verifications

	current usage for non-pyrolysis activity, as the changes in operations at the facility have been significant and usage may be much different.	conservative estimate that can be used going forward.		
Forward Action Request (Prior)	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing. Has this been completed yet?	Please provide AECN testing report. Note, likely not applicable for this reporting period, but shouldbe updated when possible	AECN has changed their plans for stack testing due to the tests no longer being required by the provincial government. They have provided an affidavit attesting that in the professional judgment of their engineering team, 1) Any residual emissions from flue gases (which are re-routed to generate heat in their biomass drier) aside from water and biogenic CO2 are de minimis, and 2) That Charm's use of the peer-reviewed stack emissions model from GHGenius ensures that any risk of understating these emissions is low	Complete

6. REVISION HISTORY

Original date of issue: June 30th, 2025

Version	Date Issued	Noted Changes
V1.0 (Draft – Internal)	June 29, 2025	N/A Initial Version
V1.1 (Draft – External)	June 30, 2025	Minor edits from QA
V1.2	July 2, 2025	Updated and closed findings summary (Section 5, Appendix 2)
V1.3 (FINAL)	July 4, 2025	Updated language regarding Isometric exception

References

[1] "Vaulted Deep Great Plains Facility Organic Waste Sequestration: Isometric PDD." Jan. 23, 2025. Accessed: Feb. 19, 2025. [Online]. Available: https://storage.googleapis.com/isometric--production--eu--public-documents/spl_1H43SYY631S0Z8YR/prj_1HHYZFVGW1S044ZY/pjd_1JJSMS9FS1S0MF1M/pdd.pdf

[2] Isometric, "Isometric Standard v1.2.0." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/standard/1.2.0

[3] Isometric, "Bio-oil Geological Storage V1.1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/bio-oil-geological-storage/1.1

[4] Isometric, "Biomass Feedstock Accounting V1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/biomass-feedstock-accounting/1.1

[5] Isometric, "Biomass or Bio-oil Storage in Salt Caverns V1.0.4." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/salt-cavern-storage/1.0

[6] Isometric, "Embodied Emissions Accounting v1.0.3." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/embodied-emissions/1.0

[7] Isometric, "Energy Use Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/energy-use-accounting/1.1

[8] Isometric, "Transportation Emissions Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/transportation

See Appendix 1 and Table 3 for list of specific files reviewed during the verification.

APPENDIX 1: DOCUMENTS SUBMITTED FOR REVIEW

File Name	File Category
Charm BO Deliveries April+May.xlsx	Charm BO MRV\Charm BO Deliveries\
Copy of Production Period 5 Summary.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\
PP5 (4_1-4_30) Biomass Throughput.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Biomass Throughput\
Miniforge Biomass Transport Allocation PP5.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Biomass Transport\
Copy of PP5 (4_1-4_30) Vendor Delivery Emissions Calc Sheet.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Consumables Delivery\
Diesel Consumption Calculator 4_1_25-4_30_25.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Diesel\
Copy of 4_17_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Diesel\Transaction Docs\
Copy of 4_28_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Diesel\Transaction Docs\
Copy of 4_7_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Diesel\Transaction Docs\
April Electricity Consumption 2025.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Electricity\
CO Grid Emissions Calc Sheet w 2025.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Electricity\
PP5 Miniforge Electricity Consumption Calculator.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Electricity\
United Power Screenshot of Date Range.png	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Electricity\
PP5 Capital Equipment MRV.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Embodied Emissions\
PP5 Gasoline Consumption Calculator.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Gasoline\
PXL_20250407_121911648.jpg	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Gasoline\Gasoline Receipts\
Production Period 5 Output Summary.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Output Data\
20250428 CAR-303.csv	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Output Data\Testing\Char\
20250428 CAR-303.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Output Data\Testing\Char\
20250421 CAR-296.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Output Data\Testing\WODO\
20250428 CAR-296_part2.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Output Data\Testing\WODO\
IMG_2295.png	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\
Propane Consumption Calculator 4_1_25-4_30_25.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\
4_11_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Bulk\
4_16_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Bulk\
4_22_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Bulk\
4_28_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Bulk\

4_7_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Bulk\
4_16_2025.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Cylinders\
4_23_2025.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Cylinders\
4_2_2025.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Cylinders\
4_30_2025.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Cylinders\
4_9_2025.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Propane\Transaction Docs\Cylinders\
4_1_25-4_30_25 Uptime Data.png	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Up Time Data\
Miniforge Pyrolysis Up-Time Reporting.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Up Time Data\
PP5 Water Usage Log.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 5\Water\
Production Period 6 Summary.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\
PP6 (5_1-5_31) Biomass Throughput.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Biomass Throughput\
Miniforge Biomass Transport Allocation -- Updated PP6.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Biomass Transport\
PP6 (5_1-5_31) Vendor Delivery Emissions Calc Sheet.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Consumables Delivery\
Diesel Consumption Calculator 5_1_25-5_31_25.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Diesel\
5_20_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Diesel\Transaction Docs\
5_5_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Diesel\Transaction Docs\
5_1_25-5_31_25 Daily Consumption Data.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Electricity\
PP6 Miniforge Electricity Consumption Calculator.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Electricity\
Screenshot of Date Range.png	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Electricity\
Cap Ex MRV May.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Embodied Emissions\
SP&C Production Period 6.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Embodied Emissions\
PP6 Gasoline Consumption Calculator.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Gasoline\
5_11_25.jpg	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Gasoline\Transaction Docs\
5_24_25.jpg	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Gasoline\Transaction Docs\
Production Period 6 Output Summary.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Output Data\
20250528 CAR-324.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Output Data\Testing 2\CHAR\
20250530 CAR-337 (1).pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Output Data\Testing 2\WODO\
Propane Consumption Calculator 5_1_25-5_31_25.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\
Copy of Agfinity 5_14.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\Transaction Docs\Bulk\

Copy of Agfinity 5_19_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\Transaction Docs\Bulk\
Copy of Agfinity 5_23_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\Transaction Docs\Bulk\
Copy of Agfinity 5_2_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\Transaction Docs\Bulk\
Copy of Agfinity 5_30_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\Transaction Docs\Bulk\
Copy of Agfinity 5_9_25.PDF	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\Transaction Docs\Bulk\
Copy of Buckeye 5_14.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\Transaction Docs\Cylinders\
Copy of Buckeye 5_7.pdf	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Propane\Transaction Docs\Cylinders\
Miniforge Pyrolysis Up-Time Reporting.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Up Time Data\
Uptime Data 5_1_25-5_31_25.png	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Up Time Data\
PP6 Water Usage Log.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 6\Water\
RP Summary April+May.xlsx	Lot Data\
Batch 3-18 #1, Batch 2-85 #1 (CHN & H2O) (2).pdf	Lot Data\2-85\
Batch 3-18 #1, Batch 2-85 #1 (TAN & Density) (2).pdf	Lot Data\2-85\
MRV Ops Notes 2-85.pdf	Lot Data\2-85\
185-10700.pdf	Lot Data\2-86\
2-86 & 3-19 Final MRV (TAN & Density).pdf	Lot Data\2-86\
2-86 Final sparged & 3-19 Post blend (CHN & H2O).pdf	Lot Data\2-86\
MRV Ops Notes 2-86.pdf	Lot Data\2-86\
185-10723.pdf	Lot Data\2-87\
185-10724.pdf	Lot Data\2-87\
2-87 & 2-88 Final MRV (TAN & Density).pdf	Lot Data\2-87\
2-87 & 2-88 Final sparged (CHN & H2O).pdf	Lot Data\2-87\
MRV Ops Notes 2-87.pdf	Lot Data\2-87\
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2-87 & 2-88 Final MRV (TAN & Density).pdf	Lot Data\2-88\
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MRV Ops Notes 2-88.pdf	Lot Data\2-88\
185-10793.pdf	Lot Data\2-89\
185-10794.pdf	Lot Data\2-89\
2-89 & 3-20 Final MRV (TAN & Density).pdf	Lot Data\2-89\
2-89 Final sparged & 3-20 Post blend (CHN & H2O).pdf	Lot Data\2-89\
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185-10874.pdf	Lot Data\2-90\
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2-90 Final sparged & batch 3-21 Post blend (CHN & H2O).pdf	Lot Data\2-90\
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185-10942.pdf	Lot Data\2-91\
185-10943.pdf	Lot Data\2-91\
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2-91 Final sparged & 3-22 Post blend (CHN & H2O).PDF	Lot Data\2-91\
MRV Ops Notes 2-91.pdf	Lot Data\2-91\
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MRV Ops Notes 2-92.pdf	Lot Data\2-92\
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185-11022.pdf	Lot Data\2-93\
2-92 & 2-93 Final MRV (TAN & Density).pdf	Lot Data\2-93\
2-92 & 2-93 Final sparged (CHN & H2O).pdf	Lot Data\2-93\
MRV Ops Notes 2-93.pdf	Lot Data\2-93\
185-11111.pdf	Lot Data\2-94\
185-11124.pdf	Lot Data\2-94\
2-94 & 3-23 Final MRV (TAN & Density).pdf	Lot Data\2-94\
2-94 Final sparged & 3-23 Post blend (CHN & H2O) (1).pdf	Lot Data\2-94\
MRV Ops Notes 2-94.pdf	Lot Data\2-94\
~~185-11112.pdf	Lot Data\2-94\
185-11198.pdf	Lot Data\2-95\
185-11199.pdf	Lot Data\2-95\
Batch 2-97 3-26 2-96 2-95 (CHN & H2O).pdf	Lot Data\2-95\
Batch 2-97 3-26 2-96 2-95 (TAN & Density).pdf	Lot Data\2-95\
MRV Ops Notes 2-95.pdf	Lot Data\2-95\
185-11219.pdf	Lot Data\2-96\
185-11220.pdf	Lot Data\2-96\
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185-11272.pdf	Lot Data\2-97\
Batch 2-97 3-26 2-96 2-95 (CHN & H2O).pdf	Lot Data\2-97\
Batch 2-97 3-26 2-96 2-95 (TAN & Density).pdf	Lot Data\2-97\
MRV Ops Notes 2-97.pdf	Lot Data\2-97\
185-11307.pdf	Lot Data\2-98\
185-11308.pdf	Lot Data\2-98\
Batch 2-98, 3-27 (CHN & H2O) Charm SR25-03586.pdf	Lot Data\2-98\
Batch 2-98, 3-27 (TAN & Density).pdf	Lot Data\2-98\

MRV Ops Notes 2-98.pdf	Lot Data\2-98\
185-10547.pdf	Lot Data\3-19\
185-10548.pdf	Lot Data\3-19\
2-86 & 3-19 Final MRV (TAN & Density).pdf	Lot Data\3-19\
2-86 Final sparged & 3-19 Post blend (CHN & H2O).pdf	Lot Data\3-19\
3-19 Charm Contents.xlsx	Lot Data\3-19\
MRV Ops Notes 3-19.pdf	Lot Data\3-19\
185-10814.pdf	Lot Data\3-20\
185-10815.pdf	Lot Data\3-20\
2-89 & 3-20 Final MRV (TAN & Density).pdf	Lot Data\3-20\
2-89 Final sparged & 3-20 Post blend (CHN & H2O).pdf	Lot Data\3-20\
3-20 Charm Contents.xlsx	Lot Data\3-20\
MRV Ops Notes 3-20.pdf	Lot Data\3-20\
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185-10897.pdf	Lot Data\3-21\
2-90 & 3-21 Final MRV (TAN & Density).pdf	Lot Data\3-21\
2-90 Final sparged & batch 3-21 Post blend (CHN & H2O).pdf	Lot Data\3-21\
3-21 Charm Contents.xlsx	Lot Data\3-21\
MRV Ops Notes 3-21.pdf	Lot Data\3-21\
2-91 & 3-22 Final MRV (TAN & Density).pdf	Lot Data\3-22\
2-91 Final sparged & 3-22 Post blend (CHN & H2O).PDF	Lot Data\3-22\
3-22 Charm Contents.xlsx	Lot Data\3-22\
MRV Ops Notes 3-22.pdf	Lot Data\3-22\
2-94 & 3-23 Final MRV (TAN & Density).pdf	Lot Data\3-23\
2-94 Final sparged & 3-23 Post blend (CHN & H2O).pdf	Lot Data\3-23\
3-23 Charm Contents.xlsx	Lot Data\3-23\
MRV Ops Notes 3-23.pdf	Lot Data\3-23\
3-24 & 3-25 (CHN & H2O).pdf	Lot Data\3-24\
3-24 & 3-35 Final MRV (TAN & Density).pdf	Lot Data\3-24\
3-24 Charm Contents.xlsx	Lot Data\3-24\
MRV Ops Notes 3-24.pdf	Lot Data\3-24\
3-24 & 3-25 (CHN & H2O).pdf	Lot Data\3-25\
3-24 & 3-25 Final MRV (TAN & Density).pdf	Lot Data\3-25\
3-25 Charm Contents.xlsx	Lot Data\3-25\
MRV Ops Notes 3-25.pdf	Lot Data\3-25\
185-11242.pdf	Lot Data\3-26\
185-11244.pdf	Lot Data\3-26\
3-26 Charm Contents.xlsx	Lot Data\3-26\
Batch 2-97 3-26 2-96 2-95 (CHN & H2O).pdf	Lot Data\3-26\
Batch 2-97 3-26 2-96 2-95 (TAN & Density).pdf	Lot Data\3-26\
MRV Ops Notes 3-26.pdf	Lot Data\3-26\

185-11328.pdf	Lot Data\3-27\
185-11329.pdf	Lot Data\3-27\
3-27 Charm Contents.xlsx	Lot Data\3-27\
Batch 2-98, 3-27 (CHN & H2O) Charm SR25-03586.pdf	Lot Data\3-27\
Batch 2-98, 3-27 (TAN & Density).pdf	Lot Data\3-27\
MRV Ops Notes 3-27.pdf	Lot Data\3-27\
185-11358.pdf	Lot Data\3-28\
185-11359.pdf	Lot Data\3-28\
3-28 Charm Contents.xlsx	Lot Data\3-28\
Batch 3-28 (CHN & H2O) Charm SR25-03638.pdf	Lot Data\3-28\
Batch 3-28 (TAN & Density) Charm SR25-03638.pdf	Lot Data\3-28\
MRV Ops Notes 3-28.pdf	Lot Data\3-28\
BOL 2167-T Pickup 5.11.pdf	Lot Data\AECN BOL_s\
Charm Trucks Pre-Commission Activity.xlsx	Site Emissions\Charm Trucks Pre-Commission Activity\
4_16_25.pdf	Site Emissions\Diesel\
5_21_25.pdf	Site Emissions\Diesel\
April_May Generator Diesel Consumption Calcs.xlsx	Site Emissions\Diesel\
252690590257.pdf	Site Emissions\Electricity\Minifinery\
Evergy bill April 30th.pdf	Site Emissions\Electricity\Minifinery\
Minifinery April_May Consolidated Energy Consumption.xlsx	Site Emissions\Electricity\Minifinery\
Vaulted Consolidated Electricity Consumption Apr-May.xlsx	Site Emissions\Electricity\Vaulted\
Copy of Copy of El Dorado to Vaulted Truck Route.png	Site Emissions\Extra Delivery\
Wild Card Emissions.xlsx	Site Emissions\Extra Delivery\
April_May Gasoline Tracker KS.xlsx	Site Emissions\Gasoline\
AB3E7B15-EB6B-4109-B205-33EA1378ADA6.jpg	Site Emissions\Gasoline\Receipts\
Copy of 1A3B6A78-FFDB-4CB6-91D3-A3DA36CF139F.jpg	Site Emissions\Gasoline\Receipts\
Copy of 26BF0785-4846-437B-8324-CE50476320AD.jpg	Site Emissions\Gasoline\Receipts\
Copy of 376FE41B-57BF-46EA-9081-B7DD571DE4E4.jpg	Site Emissions\Gasoline\Receipts\
Copy of 75716D50-8F22-45D8-910B-391227538483.jpg	Site Emissions\Gasoline\Receipts\
Copy of 8D538F75-6868-4C57-B1B5-B80D53B15A1F.jpg	Site Emissions\Gasoline\Receipts\
Copy of A8CB92D8-EF60-409A-8B47-38C958B043DC.jpg	Site Emissions\Gasoline\Receipts\
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0052948451.PDF	Site Emissions\LCS\
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April_May LCS Consumption Sheet.xlsx	Site Emissions\LCS\
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4-7.jpg	Site Emissions\Salt Deliveries\
Charm Salt Delivery Calc Sheet April_May.xlsx	Site Emissions\Salt Deliveries\
Invoice38002.pdf	Site Emissions\Salt Deliveries\
Invoice38096.pdf	Site Emissions\Salt Deliveries\
Railcar Disposal Sheet - GATX 032955.xlsx	Site Emissions\Waste Disposal\Railcar Disposal\
4_10_25.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\
4_11_25.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\
4_8_25.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\
4_9_25.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\
5_21_25.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\
5_7_25.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\
Tank Cleanouts Calc Sheet April_May.xlsx	Site Emissions\Waste Disposal\Tank Cleanouts\
AR-09 Displaced Brine 4-29-25 (1).pdf	Vaulted MRV\Brine TOC Reports\
AR-09 KOLAR April 2025.pdf	Vaulted MRV\Vaulted Reports\
AR-09 KOLAR May 2025.pdf	Vaulted MRV\Vaulted Reports\

APPENDIX 2: VERIFICATION AUDIT FINDINGS AND RESPONSES

ID No	Type	Finding / Issue	Audit Evidence	Required Action	Supplier Response	Conclusion / Resolution
1	Forward Action Request	350 is aware that Vaulted is completing some equipment and facility upgrades. If applicable to Charm injection processes, the embodied emissions should be included in Charm LCA.	NA	Please verify if Vaulted has implemented any equipment upgrades and additional equipment related to Charm's injections and ensure the Vaulted embodied emissions factor is updated if so.	Understood, will confirm with Vaulted prior to the next reporting	Review at future verifications
2	Clarification	The Kansas Grid electricity emission factor for the residual electricity mix (non-REC) for site emission 516 - removal ending in 8W94 - is indicated as 0.51089 kgCO2/kwh in Certify, but the reference source provided (PP5 Summary file) indicates an emission factor of 0.4572 for Kansas grid electricity. Can you please clarify and provide the proper updated source in Certify if this factor (0.51089) is correct or update the factor in certify if not? Note that the EF table in the PP6 summary was updated to reflect the new EF (0.51089) and we believe the correct factor is being used.	Copy of Production Period 5 Summary (https://docs.google.com/spreadsheets/d/1NBBZBb2WZsn3qa9LrYtrlfj2-0yuFk_8/edit?gid=734279272#gid=734279272)	Please clarify and provide the proper updated source in Certify if this factor (0.51089) is correct or update the factor in certify if not	The factor in Certify is correct. The value in the PP5 summary was an outdated EF for KS grid emissions. However, because the production summary only reflects CO activities, it was not used in any calculations. The value in Certify corresponds to the correct updated emission factor for electricity consumption by KS operations	Complete. No changes required.
3	Clarification	With the dedicated tanker trucks now being utilized, is Charm utilizing cleaning services to return trucks clean and empty? If so, cleaning emissions may need to be included in transport emission calculations.	April_May 2025 GHG Statement Report (https://drive.google.com/drive/u/0/folders/1z66CKgPC3_AVAVw-t72klZz93NoxrZqQ)	Please clarify	At the moment the only extra cleaning step being used for the Charm trucks is the same tank cleaning process at AJ's that third-party trucks undergo. That is included in the reporting for this period and will continue to be reported in the same way unless operations change	Complete. No changes required.

4	Forward Action Request	Non-pyrolysis electricity consumption (G&A, Eng) is based on records from May 2024. Please evaluate current usage for non-pyrolysis activity, as the changes in operations at the facility have been significant and usage may be much different.	Copy of Production Period 5 Summary.xlsx	Please update values in next Reporting Period or provide evidence that this is a conservative estimate that can be used going forward.	Absolutely, we will revisit this during the next Reporting Period	Review at future verifications
5	Omission	As stated in Verifier Guidance document, Batch 2-86 electricity usage at Vaulted was noted as incorrect in Vaulted records and was corrected based on photo evidence provided to Charm by Vaulted	Verifier Guidance -- April_May 2025	Please provide the photo evidence for the Vaulted meter	Photos have been provided in the "Verification Feedback" folder in the shared Drive	Complete
6	Recommendation	It would be preferable to share the 'MRV Ops Notes…' files as spreadsheets instead of PDFs so verifiers can evaluate calculations more easily		Provide MRV Ops files in spreadsheet form in future reporting periods	Happy to update	Complete
7	Forward Action Request	There have been some instances of significantly higher C content than typically previously seen (>50%) in some batches now. This may just be a sign of process improvements and achieving higher C content. However, please review batch carbon content history to ensure these are not outliers that could be the result of sample collection, mixing, or other systemic issues and that they are representative. If such samples are outliers, Charm could employ the outlier requirements in Section 8..2.1.1.1 of Protocol to address this as a conservative approach	Verifier Guidance -- April_May 2025	None - Completed and implemented		Complete - Charm has implemented review and we recommend continuing this practice
8	Clarification	For Removal 830, in Certify the Mass of injectant source is the MRV Ops Notes for Batch 2086, but the same source is copied three times. This occurs for every source for every removal related to the mass of injectant.	Isometric Certify Platform	Please clarify if there is a reason for this or were any other sources supposed to be uploaded? If not, please discuss with Isometric to rectify on future reporting periods	I've reviewed this, it looks like some kind of technical issue. I am not sure why the same document would be reproduced 3 times. I'll make this a ticket for our technical team to review with Isometric's	Complete

9	Omission / Forward Action Request	Lack of data obtained for salt deliveries as noted in Charms Verifier Guidance document provides less certainty in associated emissions and is an indicator of potential data tracking and management issues. Please work with suppliers to ensure proper weigh tickets / BOL are provided for each delivery to ensure accuracy. It is noted that the approach taken to fill data gaps appears conservative and overall does not materially effect the calculations of net removals. However, it does not instill confidence.	Verifier Guidance -- April_May 2025	Current approach for data gap filling is acceptable. Please consider improvements to data collection.	We will continue working with the team on site to better manage delivery drivers and improve scaling consistency	Complete for current verification - conservative approach utilized. Review at future verifications
10	Non-Compliance (Omission)	Please provide a copy of the next quarterly Report to KDHE submitted by Vaulted for Q12025 (emplaced material level in cavern). This is not required for the current reporting period, as it would be due in the next reporting period, with a due date of April 28.	NA	Submit report at next verification	Vaulted Q12025 has been submitted	Complete
11	Non-Compliance (Omission)	The Monthly Vaulted KOLAR report to KDHE for May 2025 was not provided		Please provide a copy of the report	A copy of the report has been received from Vaulted and submitted	Complete
12	Non-Compliance (Omission)	The May TOC sample results for displaced brine was not provided		Charm has indicated that the sample results are taken at end of month and may not be available until after the GHG Statement is submitted and will be submitted in next RP. 350 to discuss with Isometric. Or, Charm can provide results if available at this time	Vaulted has communicated that they were unable to perform scheduled testing due to site construction leaving wells inaccessible.	Complete – Isometric exception provided for this reporting period (Isometric email June 30, 2025 D. Lankes to K. Inder Nesbitt)

13	Recommendation (Prior)	To improve stakeholder outreach and community engagement documentation, provide documentation of attendance lists for events as well as lists of questions and answers to demonstrate the level of engagement		Provide additional supporting information if possible	Additional records have been added to the "Community Engagement Documentation" subfolder of the "Verification Feedback" folder in the shared drive. This includes a list of events and attendance and an example of more detailed notes from a recent Ft Lupton community engagement event.	Not Required. Provide updates if any available.
14	Recommendation (Prior)	Provide an improved mechanism for community complaints or inquiries. For example - provide a specific link on the Contact Us or Community Benefits web pages for 'Inquiries or Complaints from the community' or similar		Provide additional supporting information if possible	This has been added to the Charm site, please see the screenshot in the "Community Engagement Documentation" subfolder of the "Verification Feedback" folder in the shared drive.	Not Required. Provide updates if any available.
15	Forward Action Request (Prior)	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing. Has this been completed yet?	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	Please provide AECN testing report. Note, likely not applicable for this reporting period, but shouldbe updated when possible	AECN has changed their plans for stack testing due to the tests no longer being required by the provincial government. They have provided an affidavit attesting that in the professional judgment of their engineering team, 1) Any residual emissions from flue gases (which are re-routed to generate heat in their biomass drier) aside from water and biogenic CO2 are de minimis, and 2) That Charm's use of the peer-reviewed stack emissions model from GHGenius ensures that any risk of understating these emissions is low	Not Required. Provide updates in future verifications if available.

APPENDIX 3: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications	**Criteria Met?**	**Evidence / Notes** *(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).*
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Isometric Standard. Note that verifications completed for Isometric are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanantly stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☑	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☑	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☑	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☑	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☑	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Isometric Registry	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Isometric for verification, or represent the parties engaged in those activities.	☑	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☑	

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. These majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Guy Ingram-Hardwick

Carbon Removal Verification Engineer, 350Solutions

EDUCATION:

MEng Materials Science and Engineering, Loughborough University, UK - 2022

EXPERIENCE SUMMARY:

Guy Ingram-Hardwick is a Materials Engineer with experience in process engineering, MRV protocol development, experimental design and life cycle analysis (LCA). At 350 Solutions, Guys efforts center on validation and verification of varied carbon removal pathways. Guy began his experience in the carbon markets directly after graduating working to develop an LCA model for Brilliant Planet, a marine based carbon removal company growing, processing and burying microalgae for carbon sequestration. Once completing the LCA model, Guy managed the third-party verification and co-authored a paper currently awaiting publication documenting the findings. Guy also co-authored the MRV methodology for Brilliant Planet before managing its adoption with carbon market registries and developing the relevant PDD documentation. Guy's Master's thesis in degradation mechanisms for biodegradable polymers resulted in another publication but also provided him experience in experimental design and execution which lead to his experience at Brilliant Planet in conducting field trials for developing novel technologies associated to the processing and storage of the microalgae, displaying the long-term permanence of the carbon removal system. Prior to his experience at Brilliant Planet, Guy worked as a process engineer at Pirelli's rubber compound manufacturing plant in Burton-on-Trent with a focus on data analytics for driving continuous improvement.

RESEARCH AND PROFESSIONAL EXPERIENCE:

January 2025 – Present: Carbon Removal Verification Engineer, 350Solutions

Verify carbon dioxide removal technologies on behalf of registries and private companies ensuring high quality and meaningful climate impact.

Jul 2022 – Dec 2025: LCA and MRV Associate, Brilliant Planet

Quantified the carbon removal efficiency of the Brilliant Planet system across a variety of engineering designs using LCA. Developed the proprietary MRV methodology and PDD as well as setting up a novel experimental design to display the permanence of the stored carbon.

July 2019 – September 2020: Process engineer, Pirelli

Completed data analytics to drive continuous improvement for increasing efficiency, safety and rubber compound quality.